

Six-Class Latent Class Model: Consumer Profiles for Used BEV Battery Attributes

		Class 2: Range-Aware, Refurbishment-Averse n = 606 • 21%			Class 3: High-Range Enthusiasts n = 485 • 17%			Class 5: Loss-Averse, Battery-Sensitive n = 661 • 23%						
	Opt-Out			-20.8***			-13.3**			-95.8***				
	Mileage (per 10k mi)			-1.8***			-2.1***			-2.9***				
	Range at yr. 3: 40-130mi (per 100mi)									4.6.				
	Range at yr. 3: 130-200mi (per 100mi)						20.8***			4.6.				
	Range at yr. 3: 200+mi (per 100mi)						25.5***			5.3**				
	Range Loss (yr.3 - yr.8): 5-12% (per %)			-0.8***			-0.9***			-2.3***				
	Range Loss (yr.3 - yr.8): 12-24% (per %)			-0.8***			-0.7***			-2.3***				
	Range Loss (yr.3 - yr.8): 24+% (per %)			-0.4*			-0.5***			-2.1***				
	Battery Refurbishment: Pack Replace			-5.8***			-0.5			-2.7*				
	Battery Refurbishment: Cell Replace			-6.1***			-0.7			-3.8**				
CLASS CHARACTERISTICS Strong range WTP (\$9.5k–\$11.8k per 100mi), all *** Refurb-averse: pack –\$5.8k***, cell –\$6.1k*** Moderate EV battery positivity 52% · Range anxiety 83% Moderate outlet access 42% · Avg. income \$86.9k BEV ownership 7.5% · EV knowledge 77% Moderate loss sensitivity · Democratic 45%					CLASS CHARACTERISTICS Highest range WTP: \$38.7k / \$20.8k / \$25.5k per 100mi (all ***) Significant degradation aversion (all ***) Highest income \$97.1k · Highest budget \$26.9k Most EV positive: 66% enviro, 36% functional Highest outlet access 58% · Lowest SUV preference 43% Best EV knowledge 81% · Most liberal/Democratic 50% Highest used BEV purchase intent 36% agree · Youngest avg. 37.5					CLASS CHARACTERISTICS Largest class 23% · Extreme opt-out aversion –\$95.8k*** Strongest degradation sensitivity: –\$2.3k/% (all ***) Refurb-averse: pack –\$2.7k*, cell –\$3.8k** Moderate range WTP (\$4.6k–\$5.3k per 100mi) Highest risk-taking 44% · High EV positivity 63% High SUV preference 61% · Avg. income \$91.3k Outlet access 46% · BEV ownership 7.5%				
		Class 1: BEV-Skeptical Resisters n = 252 • 9%			Class 4: Budget-Constrained, Price-Dominant n = 373 • 13%			Class 6: Opt-Out Extreme, Uncertain n = 539 • 18%						
	Opt-Out			-17.6.			-34.5***			-242.4** ¶				
	Mileage (per 10k mi)			-3.2.			-0.6***			-10.0**				
	Range at yr. 3: 40-130mi (per 100mi)									12.3				
	Range at yr. 3: 130-200mi (per 100mi)													
	Range at yr. 3: 200+mi (per 100mi)									13.8.				
	Range Loss (yr.3 - yr.8): 5-12% (per %)			-1.6*			-0.2**			0.3				
	Range Loss (yr.3 - yr.8): 12-24% (per %)			-0.3			-0.2***			-1.3.				
	Range Loss (yr.3 - yr.8): 24+% (per %)			0.2			-0.2***			-1.4*				
	Battery Refurbishment: Pack Replace			-6.4*			0.1			-26.6**				
	Battery Refurbishment: Cell Replace			-7.3*			0.3			-32.6**				
CLASS CHARACTERISTICS Oldest avg. age 47 · Highest range anxiety 87% Most EV battery negative (30.5% enviro positive, 9.1% functional) Lowest outlet access 33% · Predominantly ICEV-only 89% Highest SUV preference 68% · Lowest BEV ownership 1.7% Lowest used BEV purchase intent (70% strongly disagree) Most conservative 33% · Most Republican 37% Lowest employment 38% · Lowest climate concern 22%					CLASS CHARACTERISTICS Lowest income \$70.0k · Lowest vehicle budget \$19.7k Very high price sensitivity ¶ small non-price WTPs Predominantly ICEV-only 87% · Lowest BEV ownership 3.4% Moderate outlet 40% · Range anxiety 78% Highest apartment rate 28% · Lowest homeowners 43% Moderate degradation sensitivity (\$0.2k/%), all *** Lowest vehicle budget and household vehicle count					CLASS CHARACTERISTICS Extreme opt-out aversion –\$242.4k** (scale-driven) Extreme refurb aversion: pack –\$26.6k**, cell –\$32.6k** Incoherent range WTP signs (–\$10.1k to +\$13.8k) Lowest range anxiety 67% · Most risk-taking 52% Highest BEV/PHEV ownership: BEV 13.5%, PHEV/HEV 25% Highest budget \$29.3k · Highest refuel frequency 5.5/mo Most diverse: white 59%, African American 22%				

WTP in \$1,000 USD. Piecewise linear range and range loss. Bar lengths capped at –\$120k for display; ¶ indicates truncation with actual value shown. Significance: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1. N = 2,916. Cars & SUVs combined.